Supplementary Material for "Distinct Geometric Signatures of Writing and Rewriting in AI-Assisted Text"

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This document is a technical reference for the main article. It gives the full versions of analyses that the main text reports in summary form. All quantities come from the released pipeline (seed 42) and are reported as MiniLM / OpenAI unless stated otherwise. As in the main text, the orientation difference refers to the orientation component $\Delta L_{20}^{\text{orient}}$, while raw ΔL_{20} is a composite diagnostic. Table S1 shows where to find the evidence for each question.

Table S1: Where to find the evidence for each question.

Question	Sections
Is the writing–rewriting difference robust?	S.II, S.XI, S.V (resampling)
Is the attribution of human-typed characters accurate?	S.I, S.X
Does sentence length explain the results?	S.I, S.XI
Is the writer's excess over the reference robust?	S.V, S.X
How does the out-of-reference rate depend on the radius?	S.IV
Do the results depend on the reference model (GPT-5)?	S.IX
Does human variation outside the reference have strong directions?	S.VII
What did the earlier single-distance analysis show?	S.I, S.III
How are the sentence rules and checks defined?	S.I (Table S5)
Does the human excess change within a session?	S.VI
Do writers who reject or edit more show a larger excess?	S.VIII

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S.I Extraction and Analysis Details

S.I.1 The direct-extraction error

Under direct concatenation, about half of the extracted human sentences contain at least one out-of-dictionary token. On inspection, many of these tokens are not rare words but abandoned keystrokes that the writer later corrected, such as `daughterrs`, `survbiiving`, and `ihe`. Correcting this error changes the estimated human–model distance. Consider story prompts in the first session third. With direct extraction, the mean human-to-model distance exceeds the model-to-model baseline by 0.060 (MiniLM) and 0.040 (OpenAI). After replay extraction, the values are 0.0251 and 0.0249. So the direct procedure roughly doubles the estimated separation. It also produces a large disagreement between the two embedding models, which replay extraction removes. What remains is a 1.8% out-of-dictionary rate that consists largely of typos the writer never corrected. These typos are genuine final-document text.

Table S2 gives the staged accounting for the reference pool. Because per-session suggestion counts are right-skewed, the session median (36 candidates) does not multiply out to the totals below. The query counts in the corpus metadata are lower than the replayed menu count. This is because automatic and initial suggestion displays appear in the logs but not in the query counter. We therefore base the accounting on the replay.

Table S2: From logged suggestion candidates to the Dismissed pool.

stage	count
dismissed candidates in the replayed logs (after dropping empty/very short candidates)	52,984
after prompt-material removal and exact-duplicate collapse	51,886
of which story / opinion	33,814 / 18,072

The genre correction works as follows. Inferring genre from prompt wording misclassifies two opinion prompts that begin with *What*, so 132 opinion sessions are assigned to the story group. Also, the same story prompt appears twice: once with a typographic apostrophe and once with an ASCII apostrophe. The two spellings split one prompt into two groups. The released metadata identify the 830 story sessions. For the remaining sessions, typographic punctuation is normalized before grouping.

S.I.2 Sentence provenance and examples

Replay applies each logged `retain`, `insert`, and `delete` operation at its recorded location in the document. After the text is segmented into sentences, the human-written share of each sentence is

$$h(s) = c_{\text{hum}}(s)/c_{\text{tot}}(s),$$

where both counts refer to surviving characters. Sentences dominated by prompt material are removed. For temporal analyses, each sentence is assigned to a session third by the median insertion event of its surviving characters.

The released metadata give the human-written share for each of the 830 story sessions but no such value for opinion sessions. We computed session-level shares from replay, weighted by characters over the retained sentences. These shares agree with the metadata values at $r = 0.99$, with a mean absolute difference of 2.1 percentage points. This check is at the session level. It does not verify where the ideas in a sentence came from.

The main text shows one example sentence from each set, with the characters that came from a suggestion marked.

S.I.3 Reference splitting and sentence-length controls

The dismissed pool is split into fit and held-out halves before equal-size sampling. Suggestions from the same menu can enter both halves. We test whether this overlap lowers the control baseline by keeping only one sentence per menu. With this restriction, the dimensional excess remains +2.2/+3.6 (15/20 and 17/20 groups positive).

Complete-sentence filtering addresses another asymmetry. Of the dismissed suggestions, 29% are fragments, compared with 1–2% of accepted and human sentences. With filtering, the excesses are +4.9/+7.3 (19/20 and 20/20 groups positive). A sentence counts as complete if it has at least three words. It must also begin with a capital letter, digit, quotation mark, or opening parenthesis. It must end with a full stop, question mark, exclamation mark, or closing quotation mark or bracket. The rule is applied to every set.

Four-word length matching balances candidate pools within bins. It does not balance exact lengths or every realized draw. The three-set match omits rewriting, while the four-set match includes all operations but retains fewer prompts and sentences (Table S4).

S.I.4 Choice of k and sample sizes

The writer's raw adjusted leakage is stable over $k \in \{10, 20, 30, 40\}$. It ranges from +0.029 to +0.038 across the two embedding models and is positive in 20 of 20 prompt groups at every k. But raw leakage mixes spectral shape and orientation, so we also computed the orientation component itself at each k (`orient_k_sweep.py`). For both writing (+0.016 to +0.026) and rewriting (+0.019 to +0.029), this component is positive in 20 of 20 groups at every k (Table S3). Their ordering depends on k: rewriting is higher at $k = 10$, 20, and 30, and writing is higher at $k = 40$. This sweep uses its own random draws, so its $k = 20$ values differ from the main analysis by up to 0.001 (for example, +0.029 rather than +0.030 for rewriting under OpenAI). So the positive orientation difference from the reference does not depend on where the subspace cut falls. The ranking of writing and rewriting does depend on it.

Table S3: Orientation component by subspace cut k (mean over 20 prompt groups; positive in 20/20 groups in every cell).

	MiniLM		OpenAI	
k	writing	rewriting	writing	rewriting
10	+.0158	+.0234	+.0181	+.0267
20	+.0193	+.0242	+.0225	+.0288
30	+.0210	+.0224	+.0250	+.0278
40	+.0217	+.0193	+.0263	+.0257

The cut $k = 20$ lies well below the control's own effective dimension (≈42 and 57 with equal-sized samples). Therefore the top-k block is a strict dominant subspace.

Table S4 lists the prompt groups and sample sizes for the main analysis variants. Within each comparison, all sets contain the same number of sentences. Rows marked length-matched also balance sentence lengths.

Table S4: Prompt groups and sample sizes for the main analysis variants. The participation ratio is capped at $n - 1$. So dimensional effects obtained at different sample sizes are not directly comparable across rows. The mixed-pool excess at $n = 12$ is a common calibration point. One-writer draws were run at both sample sizes shown.

Variant	Groups	n per set
main analysis	20	141–200
length-matched, four sets	10	40–62
length-matched, three sets	18	44–145
session-capped	7	41–45
complete-sentence	20	126–200
menu-capped	20	110–200
sibling-free	20	99–200
near-duplicate collapse (0.95)	20	135–200
paraphrased	20	141–200
GPT-5-anchored	20	117–120
other representations (T5, GPT-2)	20	141–200
intrinsic estimators; radius sweep	20	141–200
within-session (146 sessions)	15	12
one-writer draws (story side)	10	8 and 12
all controls stacked (145–152 sess.)	15–16	10

S.I.5 Definitions used in the checks

Table S5 defines the procedures that later sections use. It is meant to let readers rebuild each check without the code. In the released code, `replay_extract.py` implements the sentence rules, and `sibling_free.py`, `check_all_controls.py`, `scalar_sensitivity.py`, `sentence_regression.py`, `other_encoders.py`, and `nonlinear_dimension.py` implement the checks, in the order of the table.

Table S5: Definitions of the sentence rules and of the checks.

Item	Definition
Sentence splitting	The replayed final text is split after ".", "!", or "?" followed by white space (a regular expression; no library). Sentences shorter than 20 characters after trimming are dropped. A sentence is removed as prompt material if more than half of its characters come from the initial prompt.
Out-of-dictionary rate	Tokens are runs of letters and apostrophes, lower-cased, with apostrophes removed; tokens of two characters or fewer are ignored. The rate is the share of tokens not in the default English word list of `pyspellchecker` 0.9.0. Filtered variants keep sentences with a rate of 0 or of at most 10%.
Sibling-free	Every dismissed candidate from a menu from which the writer inserted a suggestion is removed from the reference.
All controls stacked	Four constraints, applied in this order: (1) complete sentences only (at least three words; first character a capital letter, digit, quotation mark, or opening bracket; last character ".", "!", "?", a closing quotation mark, or a closing bracket); (2) near-duplicates collapsed at cosine similarity ≥ 0.97 within each prompt group and set; (3) sessions with at least 10 human and 44 dismissed sentences; (4) each of 10 sampled human sentences paired with the unused dismissed sentence of the same session closest in word count. The reference is fitted on up to 120 of the remaining dismissed sentences of that session. There are 20 draws per session, and the test is a one-sample t-test over prompt-group means.
Scalar benchmark (64 variants)	Distance is the mean cosine distance from a sentence to its 5 nearest neighbours in a pool of 50 model sentences (accepted and dismissed). A linear mixed model, fitted separately for each genre, is *distance* $\sim$ source $\times$ session part + prompt, with a random intercept for each session. The 64 variants are 16 settings $\times$ 2 genres $\times$ 2 embeddings. The 16 settings are $k = 1, 3, 5, 10, 20$; pool size 20 or 100; minimum length 40 or 80 characters; pool of accepted only or dismissed only; out-of-dictionary rate 0 or at most 10%; word count as a covariate; at most 150 sentences per cell; and length matching in 5-word bins.
Sentence-level regression	The outcome is the share of a sentence's centred variance outside the top 20 principal directions of the reference. In each prompt group with at least 200 dismissed sentences, a random half of them is the reference and the other half is the control. The model is *outcome* $\sim$ source + word count, with a random intercept for each prompt. Two further fits use prompt fixed effects with standard errors clustered by session, or by writer for story prompts (57 writers).
T5 and GPT-2	`t5-base` (encoder) and `gpt2`: last hidden layer, mean pooling over tokens (at most 64), then L2 normalization.
TwoNN and Levina–Bickel	Euclidean distances between unit-length embeddings. TwoNN uses the maximum-likelihood form $N/\sum_i \log(r_{2,i}/r_{1,i})$. Levina–Bickel uses $k = 10$ neighbours, averaged as proposed by MacKay and Ghahramani. Sample size is $n = \min(200, \lfloor \lvert R\rvert/4 \rfloor, \min_c \lvert X_c\rvert)$; groups with $n < 60$ are skipped; 20 draws.

S.II Checks of the Writing–Rewriting Contrast

This section reports the checks of the primary contrast: the difference in effective dimension between writing and rewriting. All checks use the saved sentence embeddings and seed 42, with 20 draws per prompt group. The released scripts `session_matched.py`, `mixed_subtypes.py`, `mixed_subtype_geometry.py`, `mixed_origins.py`, `pr_bias_corrected.py`, and `wr_checks_summary.py` reproduce every value in Table S6. Several rows are recomputed from the prompt-level estimates of the corresponding checks in Section S.V, so these rows add the rewriting side of those analyses. They are the rows for near-duplicates, one sentence per menu, complete sentences, TwoNN, Levina–Bickel, genres, and the union dimension.

Table S6: Checks of the writing–rewriting contrast. Each entry gives the mean difference in effective dimension across prompt groups, with its 95% t interval. Then it gives the unadjusted paired t-test p and the number of positive groups. The last two blocks report the subtype and origin comparisons.

Check	MiniLM	p	Pos.	OpenAI	p	Pos.
Main samples	$+7.08$ $[+5.20, +8.95]$	2.1×10^{-7}	20/20	$+9.74$ $[+7.47, +12.00]$	2.8×10^{-8}	20/20
Story prompts	$+7.36$ $[+3.73, +10.99]$	.0013	10/10	$+9.57$ $[+5.44, +13.69]$	5.3×10^{-4}	10/10
Opinion prompts	$+6.79$ $[+4.76, +8.81]$	3.4×10^{-5}	10/10	$+9.91$ $[+7.04, +12.77]$	2.7×10^{-5}	10/10
Near-duplicates removed (0.95)	$+6.25$ $[+4.72, +7.78]$	6.2×10^{-8}	20/20	$+9.09$ $[+6.92, +11.26]$	4.2×10^{-8}	20/20
One sentence per menu	$+6.69$ $[+4.88, +8.50]$	2.7×10^{-7}	20/20	$+9.16$ $[+7.02, +11.29]$	2.9×10^{-8}	20/20
Complete sentences only	$+5.44$ $[+3.46, +7.43]$	1.6×10^{-5}	19/20	$+6.84$ $[+4.34, +9.33]$	1.6×10^{-5}	19/20
TwoNN	$+3.93$ $[+2.79, +5.07]$	7.7×10^{-7}	20/20	$+4.50$ $[+3.42, +5.58]$	4.5×10^{-8}	19/20
Levina–Bickel	$+3.67$ $[+2.93, +4.40]$	2.9×10^{-9}	20/20	$+5.33$ $[+4.57, +6.08]$	6.9×10^{-12}	20/20
Bias-corrected PR, main n	$+12.31$ $[+8.77, +15.84]$	6.5×10^{-7}	20/20	$+20.96$ $[+16.10, +25.82]$	2.7×10^{-8}	20/20
Bias-corrected PR, $n = 50$	$+11.77$ $[+8.12, +15.42]$	1.9×10^{-6}	20/20	$+20.63$ $[+15.78, +25.49]$	3.3×10^{-8}	20/20
Plug-in PR, $n = 50$	$+2.70$ $[+1.93, +3.47]$	6.0×10^{-7}	20/20	$+2.97$ $[+2.25, +3.70]$	5.7×10^{-8}	20/20
Same sessions, one each	$+4.46$ $[+2.83, +6.10]$	1.7×10^{-5}	19/20	$+5.25$ $[+3.44, +7.07]$	7.9×10^{-6}	19/20
Same sessions, two each	$+3.85$ $[+2.25, +5.46]$	7.7×10^{-5}	20/20	$+4.49$ $[+2.63, +6.36]$	7.1×10^{-5}	17/20
Writing − human–suggestion	$+2.99$ $[+1.86, +4.13]$	2.5×10^{-5}	18/20	$+3.31$ $[+2.01, +4.62]$	3.8×10^{-5}	18/20
Writing − interleaved	$+2.90$ $[+2.04, +3.75]$	1.0×10^{-6}	19/20	$+3.15$ $[+2.19, +4.11]$	1.5×10^{-6}	19/20
Writing − suggestion–human	$+2.91$ $[+1.83, +4.00]$	6.2×10^{-5}	14/14	$+2.78$ $[+1.83, +3.73]$	2.7×10^{-5}	14/14
Interleaved − human–suggestion	-1.13 $[-2.05, -0.21]$	.019	6/20	-1.27 $[-2.29, -0.26]$	.017	4/20
Selection: originals − reference	$+0.22$ $[-1.18, +1.63]$	.74	12/20	-0.12 $[-1.51, +1.28]$	.86	9/20
Editing: final − originals	-3.07 $[-4.76, -1.37]$	.0013	3/20	-2.56 $[-4.55, -0.56]$	.015	4/20
Union with reference	$+4.93$ $[+3.69, +6.17]$	9.0×10^{-8}	20/20	$+7.31$ $[+5.69, +8.93]$	1.4×10^{-8}	20/20

Same sessions. This check holds the sessions fixed. Within each prompt group, we keep the sessions that have at least m Mixed, m Human, and m Dismissed sentences. From every such session, we draw m sentences of each type. This gives 42–82 sessions per group for $m = 1$ and 26–66 sessions for $m = 2$, so the three samples have identical session composition.

Rewriting subtypes. Replay gives the author of every surviving character, in order, so we collapse the author sequence of each Mixed sentence into runs. We ignore runs without word characters. The resulting patterns define the subtypes. Human text followed by suggestion text is *human–suggestion* (1,966 sentences, 46%). Suggestion text followed by human text is *suggestion–human* (799, 19%). Three or more alternating runs are *interleaved* (1,159, 27%). The remaining 350 sentences (8%) fit none of these. These names describe the order of characters in the final sentence, not the order in which they were typed. The median human-typed shares of the three subtypes are 0.46, 0.44, and 0.52, respectively. For each subtype with at least 25 sentences in a group, we compare the subtype, Human sentences, and held-out Dismissed sentences. The three samples have equal size ($n = \min(|S|, \lfloor |R|/4 \rfloor, 200)$). Suggestion–human has enough sentences in 14 groups.

Original suggestions. We can recover the original suggestion behind each Mixed sentence. The surviving suggestion characters of a Mixed sentence carry the index of the event that inserted them. So the string inserted by that event is the original suggestion. We join the strings of all such events in order, which recovers originals for 3,986 of 4,274 Mixed sentences. Of these, 479 draw on more than one event, so their "original" may combine more than one suggestion. We also restrict the analysis to the 3,507 sentences whose surviving suggestion text comes from a single event. This gives the same pattern: selection $+0.75$ / $+0.29$ ($p = .22$ / .67) and editing -3.31 / -2.76 (3/20 and 5/20 groups positive). Median lengths are 20 words for the final sentences and 11 for the originals. We embed the originals with both models and then keep one sentence per distinct original within each prompt group. *Selection* compares the originals with held-out Dismissed sentences at equal n (134–200). Both sets are unedited model suggestions. *Editing* compares the final sentences with their own originals, on the same sampled sentences.

Bias-corrected participation ratio. Let S be the sample covariance. Then $\hat{a}_2 = \frac{(n-1)^2}{(n-2)(n+1)}\big[\operatorname{tr}(S^2) - (\operatorname{tr} S)^2/(n-1)\big]$ is unbiased for $\operatorname{tr}(\Sigma^2)$ under normality (M. S. Srivastava, "Some tests concerning the covariance matrix in high dimensional data," *Journal of the Japan Statistical Society*, vol. 35, no. 2, pp. 251–272, 2005, doi:10.14490/jjss.35.251). Also, $(\operatorname{tr} S)^2 - 2\hat{a}_2/(n-1)$ is unbiased for $(\operatorname{tr}\Sigma)^2$. The ratio of these two estimators

is the corrected participation ratio. On the same draws, the plug-in reference dimension is 42.1/57.0 at the main sizes and 25.4/30.4 at $n = 50$. But the corrected values are 55.8/84.9 and 56.3/85.2. Embeddings are not Gaussian and sentences within a document are dependent, so the correction is approximate.

Other values cited in the main text. The main text cites two more comparisons. In the four-set length-matched samples ($n = 40$–62), writing exceeds the held-out reference by +0.83 and +0.78 (7/10 and 8/10 groups). The reference dimensions are 24.3 (MiniLM) and 29.1 (OpenAI). In the same-session comparison with one sentence from each session, writing exceeds the reference by +1.62 and +2.01 (15/20 and 18/20 groups).

Orientation component. Each bracket of the orientation component is nonnegative. This follows from the Eckart–Young theorem: no k-dimensional subspace captures more of a set's variance than the set's own top k principal directions. In the main samples, the top 20 principal directions of a reference sample capture on average 55% and 47% of its variance.

Holm family. The Holm family contains six tests. They are the writing–rewriting contrast and the writing-versus-reference tests of dimension and orientation, each under both embedding models. For this contrast, the adjusted p-values are 6.2×10^{-7} and 1.1×10^{-7}. They are .00109 and .00091 for the writing–reference dimension and 2.1×10^{-11} and 2.6×10^{-11} for orientation.

S.III The Scalar Benchmark

Table S7 gives the full scalar result, which the main article summarizes in the prior-work comparison in its Related Work. A mixed model with a session-level random intercept estimates the human-to-model gap and how this gap changes during the session.

Table S7: The scalar gap ($d_{\mathrm{H}\to\mathrm{L}} - d_{\mathrm{L}\to\mathrm{L}}$) and its change over the session, by genre and embedding model (MiniLM / OpenAI).

	story	opinion
gap, first third	+0.0251 / +0.0249	+0.0351 / +0.0387
	p = 2.2e-41 / 1.4e-59	p = 8.1e-26 / 3.2e-45
change, first to last	−0.0120 / −0.0152	−0.0075 / −0.0105
	p = 5.8e-6 / 2.1e-12	p = 0.096 / 0.005

The gap is positive at $p < .001$ in 64 of 64 variants of the scalar measure (Table S5 defines them). The narrowing has the same sign in 64 of 64 variants. It reaches $p < .05$ in 30 of 32 story tests and 13 of 32 opinion tests. Consider first the inferred genre labels, which assigned 132 opinion sessions to the story group. Under these labels, the source × session-position × genre interaction reached $p = .063$ and $p = .016$. The opinion prompts showed no narrowing at all ($p = .57$ and $p = .19$). With the corrected metadata labels, by contrast, the interaction does not reject the null ($p = .37$ and $p = .23$). The correction leaves every other result nearly unchanged, most of them to three decimal places. However, once the genre labels are corrected, the previously reported genre difference is no longer statistically supported.

S.IV Out-of-Reference Rate: Radius Sweep

Table S8 reports the out-of-reference rate at each radius multiplier (main-text Results). At every radius up to 2.0×, the ordering of conditions stays the same: writing lies above the control, while choosing and rewriting lie below it. The radius sets the magnitude, not the ranking. From 2.0× upward, the rates of all conditions become very small.

Table S8: Out-of-reference rate by radius multiplier (MiniLM). The table gives the control's own rate and the excess of each condition over it. The OpenAI embedding model gives the same pattern.

radius	control	Choose	Rewrite	Write
1.0×	.494	−.034	−.073	+.082
1.5×	.087	−.025	−.037	+.013
2.0×	.010	−.004	−.006	+.002
2.5×	.001	−.000	−.001	+.000

S.V Full Robustness Inventory

Table S9 lists the checks of the writer's excess over the reference. The checks are grouped by the question they address, and later subsections give the details. Section S.X reports the process and proxy-matching analyses. Section S.II reports the checks of the writing–rewriting contrast.

Table S9: Checks of the writer's excess over the reference (MiniLM / OpenAI), grouped by the question they address. This is the full version of the writing–reference column of the main text's table of checks. Dimensional entries give the excess, with positive prompt groups in parentheses. Raw-leakage entries combine spectral shape and orientation. Null-calibration rows give the largest value in each analysis's null runs, and the stacked directional null carries a small negative offset. For saturation, the row gives the trend across sizes. Designs and sample sizes differ across rows, and the participation ratio is capped at $n - 1$. So compare magnitudes only within a row.

Check	Explanation tested	Dimensional excess	Raw leakage (composite)	Unit	Source
Are the texts different in other ways?					
Length matched, three sets	sentence length	+2.15 / +2.63 (16/18, 16/18)	+.031 / +.032 (18/18)	18 groups	S.V
Length matched, four sets	length, all sets	+0.83 / +0.78 (7/10, 8/10)	+.027 / +.024 (10/10)	10 groups	S.XI, S.II
Complete sentences only	suggestion fragments	+4.9 / +7.3 (19/20, 20/20)	—	20 groups	S.I
One sentence per menu	repeated candidates	+2.2 / +3.6 (15/20, 17/20)	+.029 / +.033 (20/20)	20 groups	S.I
Session capped at two sentences	document clustering	+0.80 (MiniLM; 6/7)	+.022 (7/7)	7 groups	S.V
Near-duplicates removed (0.95)	repeated or paraphrased text	+1.06 / +2.7 (12/20, 16/20)	+.025 / +.030 (20/20)	20 groups	S.V
Position matched (W=50)	position in the document	+3.27 / +3.63 (20/20, 19/20)	—	20 groups	S.V
Similarity and perplexity matched	off-topic or disfluent text	+1.64 / +3.98 (16/19, 17/17)	—	19 / 17 groups	S.X
Does the classification of sentences matter?					
Fully human-typed ($h = 1$)	authorship threshold	+2.72 / +4.56 (17/20, 18/20)	—	20 groups	S.V
Process-filtered human	overwritten or retyped suggestions	+2.75 / +4.37 (17/20, 19/20)	—	20 groups	S.X
Does the result depend on the estimator or the representation?					
TwoNN estimator	linear-estimator dependence	+5.6 / +7.3 (20/20, 20/20)	—	20 groups	S.V
Levina–Bickel estimator	linear-estimator dependence	+3.0 / +4.2 (19/20, 19/20)	—	20 groups	S.V
Saturation, $n = 25$–400	small-sample bias	gap grows to +2.7 / +4.8	—	17 groups	S.V
T5 encoder states	contrastive training	+4.9 (19/20)	+.046 (20/20)	20 groups	S.V
GPT-2 decoder states	contrastive training	not resolvable	+.025 (20/20)	20 groups	S.V
Paraphrased into one register	register and surface form	+1.2 / +2.4 (12/20, 14/20)	+.021 / +.025 (20/20)	20 groups	S.V
Does writing change the combined collection?					
Union with reference	effect on the combined collection	+2.5 / +4.6 (19/20, 20/20)	—	20 groups	S.V
Is the effect due to dependence among sentences?					
Within one session ($n = 12$)	writer and context mixing	+0.6 / +0.5 (15/15, 12/15)	+.095 / +.095	146 sessions (15 groups)	S.V
All controls stacked ($n = 10$)	combined sampling artifacts	+0.37 / +0.57 (11/15, 13/16)	+.055 / +.096	145 / 152 sessions (15 / 16 groups)	S.V
Writer jackknife (stories)	writer as inference unit	+2.8 / +3.6, wide CIs	—	57 writers	S.V
Writers and prompts resampled (distinct sentences)	shared writers	+3.43 / +4.33, intervals exclude zero	—	57 writers, 10 prompts	S.V
Sentence-level regression	length as covariate; writer as unit	—	+.043 / +.045	58,617 sentences	S.V
Null calibrations					
Prompt groups	estimation noise	null ≤ 0.59	null ≤ 0.0023	200 / 100 runs	S.V
Within one session	estimation noise at n=12	null $\leq .071$ / .068	null $\leq .0043$ / .0048	30 runs	S.V
All controls stacked	estimation noise at n=10	null $\leq .091$ / .115	offset $-.024$ / $-.023$	30 runs	S.V
For comparison: does choosing add dimensions?					
Same-menu selection (choosing)	unmatched suggestion contexts	-2.12 / -3.38 (4/20, 1/20)	—	20 groups	S.V

Notes. The three-set length match covers the reference, choosing, and writing. By contrast, the four-set match and the main text's table of direct contrasts also include rewriting. "Process-filtered" keeps human sentences with no logged trace of suggestion use (Section S.X). The within-session and stacked rows use very small samples, so read them against their own null calibrations. The resampling row is the weighted-distinct variant. The standard bootstrap intervals include zero. The last row reports the choosing effect, not the writing effect.

S.V.1 Null calibrations

The two measures exceed the largest null value by different margins. The raw-leakage excess exceeds it by an order of magnitude, while the dimensional excess exceeds it by a factor of four in the main analysis. Under the most conservative length-matched variant, however, the dimensional excess comes close to that value. So the dimensional claim rests on consistency across analyses, not on any single conservative subset.

We test estimation noise separately in the within-session and combined-control analyses. In the within-session analysis, the null calibration replaces the writer's sentences with a second draw of the session's own rejected suggestions. We rerun the full pipeline on this null, with 30 replications under each embedding model. The null excess never exceeds 0.071 and 0.068 in dimension, or 0.0043 and 0.0048 in direction. So the observed dimensional excess is 8.2 and 6.8 times the largest of the 30 null values of its embedding model. It is 17 and 14 null standard deviations from the null mean. The raw-leakage excess is 22 and 20 times the largest null value. These multiples describe size relative to the null runs; they are not significance tests. In the combined-control analysis, the dimensional null never exceeds 0.091 and 0.115. So the observed +0.37 and +0.57 are 4.0 and 5.0 times the corresponding largest null value. The directional null of this analysis lies slightly below zero (-0.024 ± 0.005 and -0.023 ± 0.003) because of the length-pairing step. The observed excess lies roughly 17 and 36 null standard deviations above this null. Session-level draws use a fixed seed derived from each session identifier.

The within-session and stacked values need a size calibration, because the participation ratio is capped at $n-1$. For this purpose, we reran the main mixed-pool analysis at n=12. This gives +0.10 and +0.09 (control dimension 9.0 of a maximum 11). So the within-session +0.6 is not the small value that the cap alone would predict. A mixed-writer comparison cannot use the same within-session sampling rule, because a mixed draw necessarily spans sessions. Instead, the null calibrations above use that rule directly.

S.V.2 Mixing writers and contexts

This check holds the reference fixed and draws the writer's sample from a single writer (median two sessions within a prompt group). Under this design, the raw-leakage excess grows rather than shrinks. One-writer draws give +0.055, against +0.036 and +0.038 for mixed draws. The excess is positive in every group under both embedding models. So mixing writers dilutes this raw-leakage excess rather than producing it.

For dimension, a one-session draw falls below the cross-context control. The reason is that the dimension of a draw tracks how many contexts it spans. This context effect can be measured. At n=12, a control draw that spans sessions has dimension 9.0 and 9.6 under the two embedding models. A draw from a single session has 7.7 and 8.3. So spanning contexts contributes about 1.3 of the control's own dimension at that size. Both pools share this structure in the group-level analysis. Neither pool has it in the within-session analysis.

At most two sentences per session. A further check limits every sample, including the reference, to at most two sentences from any one session. This reduces the dependence among sentences from the same document. It also shrinks every pool, so rewriting is left out. Only seven prompt groups keep enough sentences ($n = 41$–45 per set). Under MiniLM, the writer's dimensional excess is +0.80 (6/7 groups positive), and the directional excess is +.022 (7/7).

S.V.3 Rewrite clustering (anchoring)

This check asks which set clusters most tightly. With equal-sized samples, rewritten sentences have the smallest within-set nearest-neighbour distances. They also have the highest share of nearest neighbours drawn from their own session. Each result holds against the control in 18 or more of 20 groups, under both embedding models. On both diagnostics, the writer's sentences show the opposite pattern: they have the least internal redundancy of any set. The main text's authorship analysis splits sentences into heavy- and light-rewrite bins. Within these bins, heavily rewritten sentences have smaller within-set nearest-neighbour distances than lightly rewritten ones (0.39 against 0.42 under MiniLM, 18/20 groups, p=3e-4; 0.43 against 0.47 under OpenAI, 17/20, p=3e-5). The two bins share one length distribution (mean 22.3 and 21.6 words). Heavy rewrites keep fewer suggestion characters, yet they cluster more. So the clustering does not increase with the amount of suggestion text retained. We observe it within the writer's documents. This check does not identify its cause.

S.V.4 Alternative estimators, sample sizes, and model families

The participation ratio summarizes how variance is spread across principal directions. TwoNN, by contrast, estimates the local dimension of a nonlinear manifold. Rewriting increases the nonlinear estimate (+1.7 and +2.8) but reduces the linear estimate. Only writing increases both estimates, a pattern that also holds under the Levina–Bickel estimator. The saturation analysis should be read as a trend, not as a series of per-step tests. Across sizes, the writer's dimensional gap grows from +0.35 / +0.45 at $n = 25$ to +2.7 / +4.8 at $n = 400$. Growth with sample size alone cannot rule out a bias that differs between sets. But the mean gap is positive at every size we examined. The bias-corrected participation ratio (Section S.II) also gives a positive gap at the main sizes (+4.00 / +8.20, 16/20 groups under both models). Under T5, a denoising encoder with no contrastive objective, the whole pattern returns. GPT-2 is a decoder whose space is too anisotropic to resolve dimension (4.1 effective directions of 768). Under this model, the directional ordering still replicates.

The Vendi score points the same way. It is an effective number of distinct items, based on the spectrum of cosine similarities. With equal-sized samples, the human sample exceeds the held-out control by +3.5 under MiniLM (control 72.9, p=7.6e-4, 15/20). Under OpenAI, it exceeds the control by +4.5 (control 95.8, p=4.1e-4, 19/20).

S.V.5 Paraphrase control

One model (GPT-5) paraphrased every sampled sentence, from both model and human sources, into the same plain register. This succeeded for 30,128 of 30,129 sentences (one failure). We re-embedded the paraphrases and reran the pipeline. Paraphrasing leaves the reference's own dimension essentially unchanged (control d_{eff} 42.7→43.4 and 58.0→58.6), so the paraphraser does not collapse the space. The writer's excess survives on both forms, while choosing stays at or below the control. A positive residual does not establish that the retained variation is semantic. The paraphraser may preserve stylistic differences or change content. The reduced dimensional excess (+1.2/+2.4) is consistent with sensitivity to wording. But it is not a causal decomposition of the original effect into semantic and stylistic fractions. Fidelity is moderate: the original-to-paraphrase cosine is 0.85–0.87. The human–model difference is below 0.004 and of inconsistent sign. Similar average similarity across sources does not establish content preservation.

S.V.6 Spectrum-aligned decomposition of raw leakage

This decomposition reduces the effect of spectral shape on leakage. A flatter spectrum leaks more variance past any fixed top-k subspace, even when orientation is unchanged. So the observed ΔL_{20} mixes spectral shape with orientation. We therefore calculate the leakage that each condition would show if its principal axes coincided, in order, with the reference's axes. This spectrum-aligned null is the tail share of the condition's own spectrum beyond k. We subtract this aligned leakage from the observed leakage for both the condition and the held-out control. The difference between these adjusted values is the orientation component (Table S10). The subtraction does not remove spectral shape completely: each misaligned direction still counts in proportion to its variance. The component is therefore a variance-weighted misalignment, not a pure direction measure. Most of the writer's raw-leakage excess falls in this component. Rewriting carries an orientation component that the raw measure cancels against its narrower spectrum. This is why its directional zero point moves with the composition of the reference.

Table S10: Decomposition of the control-adjusted leakage ΔL_{20} into a spectral-flatness component and an orientation component (MiniLM / OpenAI). For the orientation component, parentheses give the number of groups positive out of 20.

	observed	flatness	orientation
Choose	−.007 / −.007	−.013 / −.016	+.006 (18) / +.009 (19)
Rewrite	−.001 / +.005	−.026 / −.025	+.025 (20) / +.030 (20)
Write	+.032 / +.035	+.012 / +.011	+.**020** (20) / +.**024** (20)

Table S11 gives the raw leakage and effective-dimension excesses by authorship bin.

Table S11: Excess over the model control by human-written share h (MiniLM, 20 prompt groups, with the same n in every bin). The two -1.45 entries are equal only by a rounding coincidence (-1.454 and -1.449). Values differ from Table S14 because the bins share a smaller sample size n.

h	raw leakage	effective dimension
< 0.2	-0.008 (p=.002)	-1.45 (p=1.5e-4)
0.2–0.5	$+0.005$ (p=.18)	-1.45 (p=.023)
0.5–0.8	-0.003 (p=.57)	-4.61 (p=5.2e-5)
≥ 0.8	$+0.027$ (p=7.2e-7)	$+1.32$ (p=.013)

The trend across bins is significant (raw leakage $+0.010$ per step, p=1.5e-7, 20/20; dimension $+0.52$ per step, p=.004, 16/20). However, the bin means do not show a steady dimensional increase. We also apply the same decomposition to the four authorship bins of the main text's authorship analysis. It gives an orientation gradient in editing depth (Table S12). The orientation component rises from the near-model bin through light to heavy rewriting. It is largest for heavy rewrites and exceeds the human bin. The human bin's raw lead comes from its flatness term.

Table S12: Spectrum-aligned decomposition by authorship share h (MiniLM / OpenAI). For the orientation component, parentheses give the number of groups positive out of 20.

h	observed	flatness	orientation
< 0.2	$-.008$ / $-.007$	$-.012$ / $-.015$	$+.005$ (18) / $+.008$ (20)
0.2–0.5	$+.005$ / $+.010$	$-.012$ / $-.008$	$+.017$ (20) / $+.018$ (20)
0.5–0.8	$-.003$ / $+.003$	$-.035$ / $-.033$	$+.032$ (20) / $+.036$ (20)
$\geq$ **0.8**	$+.027$ / $+.030$	$+.010$ / $+.011$	$+.017$ (20) / $+.019$ (20)

S.V.7 A decomposition-free orientation check: principal angles

The orientation component subtracts the leakage expected when the principal axes align with the reference. We also compare the principal directions directly, without this subtraction. For this, we use the mean squared cosine of the principal angles between each condition's top-20 principal directions and the reference top-20 subspace. Formally, the unadjusted misalignment is

$$1 - \|P_k V_X^\top\|_F^2 / m,$$

where P_k is the reference projector and the rows of V_X are the top $m = 20$ orthonormal principal directions of X. The unadjusted value lies in $[0, 1]$. Subtracting the held-out control gives the reported excess. Over the held-out control, the misalignment excess is $+.0031$ / $+.0039$ for choosing (n.s.). It is $+.0309$ / $+.0345$ for rewriting (19–20/20, $p \leq$2.3e-8) and $+.0421$ / $+.0523$ for writing (20/20, $p \leq$9.2e-12). The two orientation measures order rewriting and writing differently. The orientation component places rewriting slightly higher, while principal angles place writing higher. Under either measure, both operations show larger orientation effects than choosing. The principal-angle statistic is not fully spectrum-free. For a flatter spectrum, the top-m directions are estimated less stably, which inflates misalignment. The flatter sets are the writer's. So the two orientation measures carry opposite spectral biases. Both measures separate writing and rewriting from the control, but the ranking of the two depends on the measure and on k (Section S.I.4).

S.V.8 How many sessions contribute to each sample?

This check counts the distinct sessions behind each sample. In the main analysis, the median session counts per equal-sized sample are 57 (control), 56 (Accepted), 51 (Mixed), and 61.5 (Human). For the four authorship bins, the medians are 42.2 / 35.8 / 39.0 / 46.5. The draws depend only on the seed, so these values are identical under both embedding models. The samples thus span many sessions, but these counts alone do not exclude within-session dependence or unequal contributions. Section S.XI reports per-prompt contribution diagnostics and a session-weighting sensitivity analysis.

S.V.9 Genre split of the dimensional effect

Here we split the main dimensional excess by genre. Under MiniLM, the excess is +2.91 for story (9/10 groups positive) and +1.68 for opinion (6/10). The five negative groups are four opinion prompts and one story prompt. With OpenAI embeddings, the excess is +3.54 for story (9/10) and +3.84 for opinion (9/10). Each genre has one negative group.

S.V.10 Additional controls for choosing and writing

Five further controls sharpen the dimensional claim.

(1) Menu-matched choosing. We compare the accepted set with only the rejected candidates from the same menus. Against these candidates, the Accepted set shows no increase in effective dimension (−2.12 / −3.38, $p \leq$1.4e-4; 4/20 and 1/20 groups positive). But it does carry a small orientation excess (+.0033 / +.0068, 16/20 and 17/20).

(2) Union dimension. We combine each condition with the reference and then measure the effective dimension of the pooled collection: $U(X) = d_{\text{eff}}(R_{\text{fit}} \cup X) - d_{\text{eff}}(R_{\text{fit}} \cup R_{\text{hold}})$. The results are Human +2.47 / +4.57 (19/20, 20/20; $p \leq$2.2e-6), Accepted −0.80 / −1.39, and Mixed −2.46 / −2.74. We also ran a centred variant. It recentres every set on its own mean before pooling, so the distance between centroids cannot contribute. This variant gives Human +2.60 / +4.76 (19/20, 20/20; $p \leq$1.2e-6), while Accepted and Mixed remain negative.

(3) Fully human-written sentences. We keep only sentences with $h = 1.0$, which make up 96% of the human set. On this subset, the results are: effective dimension +2.72 / +4.56 (17/20, 18/20), orientation +.0203 / +.0243 (20/20), and raw +.0343 / +.0373 (20/20).

(4) Choosing with no human-typed characters. Accepted sentences are analyzed in their final-document form. In this form, they may contain less than 20% human-typed characters. Of these sentences, 81% contain none. The mean human-typed share is 1.7%. We restricted the Accepted condition to sentences with no human-typed characters, using equal-sized samples at $n \leq 200$. This restriction reproduces the choosing results. Under MiniLM and OpenAI, respectively, the dimensional excess is −1.91 and −3.41 (4/20 and 1/20 groups positive, $p \leq .0018$). The orientation excess is +0.008 and +0.011 (20/20 and 19/20, $p \leq 2 \times 10^{-7}$).

(5) Document-position control. Model suggestions appear only at the points where the writer requests them, but human sentences come from the whole document. For every sentence, we computed two values. The first is the median insertion-event index of its surviving characters. The second is its distance, in events, to the nearest suggestion-open event in the same session. For human sentences, the median distance is 89 events. We then kept only human sentences within W events of a request. On these sentences, we reran the dimensional and orientation contrasts (equal-sized samples; $n \leq 100$, minimum 40). Under MiniLM / OpenAI, the dimensional excess is +3.58 / +3.45 at W=25, where 8% of human sentences are kept (15 groups). At W=50 (28% kept, all 20 groups), it is +3.27 / +3.63 (20/20 and 19/20, $p \leq 1.5 \times 10^{-5}$). At W=100 (54% kept), it is +2.04 / +2.83. The orientation excess remains positive in every group at every radius (+0.014 to +0.023). So the dimensional excess remains positive when human sentences are restricted to positions near suggestion requests.

S.V.11 Resampling writers and prompts

This check resamples writers and prompts together. A two-way cluster bootstrap resamples the 57 story-side writers and the ten story prompts. With five sampling replications per prompt draw, the dimensional analysis uses $B = 2{,}000$ draws under each embedding model (seed 42). The bootstrap means are +2.40 under MiniLM (percentile 95% CI $[-0.18, +4.99]$) and +2.58 under OpenAI ($[-0.91, +6.02]$). Both intervals include zero. The fractions of nonpositive draws are .0325 and .0675. These are descriptive bootstrap fractions, not null-calibrated p-values.

The intervals depend little on the number of draws above 1,000. At $B = 1{,}000$, they are $[-0.18, +4.92]$ and $[-0.96, +6.04]$. Each endpoint changes by less than 0.07 effective dimensions between $B = 1{,}000$ and $B = 2{,}000$. With only 200 draws, however, the MiniLM interval is $[+0.03, +4.96]$. Its exclusion of zero is therefore not stable as the number of draws grows. Uncertainty about generalization across writers remains under both embedding models. For computation, we evaluate d_{eff} from the centred Gram matrix K as $(\operatorname{tr} K)^2 / \sum_{ij} K_{ij}^2$, which is algebraically equivalent to the spectral formula. The two computations agree to within 5×10^{-6}.

Directional estimates use $B = 200$ for MiniLM and $B = 100$ for OpenAI. Under MiniLM, raw leakage is $+0.045$ $[+0.031, +0.061]$ and orientation is $+0.036$ $[+0.027, +0.047]$. The corresponding OpenAI values are $+0.045$ $[+0.031, +0.063]$ and $+0.039$ $[+0.028, +0.051]$.

We also ran a writer-cluster bootstrap. It resamples the 57 story-side writers with replacement (B=300 MiniLM / B=150 OpenAI, 5 pipeline replications per draw). Under MiniLM, the story-side dimensional excess is +2.46, with 95% CI $[+0.48, +4.36]$ and $P(\leq 0) = .007$. Under OpenAI, it is +2.61, with CI $[-0.02, +5.42]$ and $P(\leq 0) = .027$. One interval is clear of zero, while the other touches it. The bootstrap raw-leakage excesses are +0.045 and +0.047, with CIs $[+0.034, +0.061]$ and $[+0.035, +0.063]$. Both are clear of zero.

A delete-one-writer jackknife over the same 57 writers gives a story-side dimensional excess of +2.8 and +3.6. The 95% CIs are $[-0.6, +6.2]$ and $[+0.03, +7.3]$. So the interval is positive under one embedding model and crosses zero under the other. Both intervals are wide, because a few high-volume writers dominate the pools. Writer-level inference for dimension is therefore weaker than the group-level results and the analysis-specific null calibrations. For direction, by contrast, the writer-clustered regression interval is narrow and clear of zero (+0.041, 95% CI $[0.034, 0.048]$). Adding the sentence's session position as a covariate moves it by less than 0.003.

Duplicate sentences in resamples. In the two-way bootstrap above, a writer drawn k times contributes k copies of each of their sentences to the candidate pools. Each geometric sample of size n is then drawn without replacement from these pools. So a sample can contain the same sentence more than once. Repeated points change the finite-sample dimension. Because smaller pools give more chances for repeat selection, this can affect the two contrasts differently. The observed shifts between procedures are consistent with this mechanism. However, they do not establish which procedure has correct bias or interval coverage. In particular, removing duplicates is not by itself a correction to the standard bootstrap. The one-way writer bootstrap reported earlier in this section has the same property, but the delete-one-writer jackknife does not.

We therefore ran two variants (code: `writer_prompt_bootstrap.py`, options `--nodup`, `--wdistinct`, `--vs-mixed`; seed 42; 1,000 draws). In *no-duplicate* resampling, every sentence of a drawn writer enters the pool once, regardless of k. This removes writer weights. It behaves like subsampling about 63% of writers, so it does not reproduce ordinary bootstrap weighting. Its interval coverage has not been established here. In *weighted distinct* resampling, each sentence enters once with weight k. Samples are then drawn without replacement with probability proportional to weight. This keeps writer weights but prevents duplicates within a sample. The variant modifies the standard bootstrap and is not an established procedure, so we report it as a sensitivity analysis. For the writing–rewriting contrast, the comparison sample comes from the rewritten pool instead of the held-out reference half. In this contrast, n is limited by the smaller pool.

Table S13: Two-way (writer × prompt) bootstrap of the effective-dimension contrast for story sessions (57 writers, 10 prompts). Entries give the mean and the percentile 95% interval. The standard human–reference row uses 2,000 draws (see above). All other rows use 1,000.

Contrast	Resampling	MiniLM	OpenAI
Write − reference	standard (duplicates)	$+2.40$ $[-0.18, +4.99]$	$+2.58$ $[-0.91, +6.02]$
	no-duplicate	$+3.40$ $[+1.26, +5.68]$	$+4.26$ $[+1.40, +7.47]$
	weighted distinct	$+3.43$ $[+0.96, +5.97]$	$+4.33$ $[+1.08, +7.59]$
Write − rewrite	standard (duplicates)	$+12.14$ $[+7.30, +17.72]$	$+17.30$ $[+10.96, +24.80]$
	weighted distinct	$+6.79$ $[+3.53, +10.50]$	$+8.53$ $[+4.73, +12.85]$

The writing–rewriting difference excludes zero under every variant and under both embedding models. Its sign is stable, but its size depends on the procedure. For example, under MiniLM it is +12.14 in the standard bootstrap and +6.79 in the weighted distinct variant. The coverage of these intervals has not been calibrated by simulation. The human–reference excess excludes zero once duplicates are removed, but it does not exclude zero in the standard bootstrap.

S.V.12 Main operation effects

Table S14 gives the numerical estimates behind the MiniLM bars in Figure 2 of the main article. The main article reports direct comparisons between operations, while the table below keeps each operation's comparison

with the held-out model control.

Table S14: Main MiniLM effects across 20 prompt groups. Each entry gives the mean excess over the held-out model control ± the half-width of the 95% confidence interval. Numbers in parentheses count groups with a positive effect.

Operation	out-of-reference rate	effective dimension	orientation difference
Choose	$-.004 \pm .004$	-1.66 ± 0.91	$+.006 \pm .002$ (18/20)
Rewrite	$-.006 \pm .007$	-4.78 ± 2.15	$+.025 \pm .005$ (20/20)
Write	$+.002 \pm .005$	$+\mathbf{2.30} \pm \mathbf{1.25}$ (15/20)	$+\mathbf{.020} \pm \mathbf{.003}$ (20/20)

S.V.13 Direct comparisons between writing operations

Table S15 adds the p-values of the direct contrasts in the main text, from two-sided paired t-tests (unadjusted). Main samples contain 141–200 sentences per set, while length-matched samples contain 40–62. So magnitudes cannot be compared directly between these analyses.

Table S15: Direct contrasts in effective dimension. "Positive" counts the prompt groups with a positive difference. CI denotes a 95% confidence interval.

Analysis	Model	Contrast	Mean [CI]	p	Positive
Main	MiniLM	Write–Rewrite	7.08 [5.20, 8.95]	2.1×10^{-7}	20/20
Main	OpenAI	Write–Rewrite	9.74 [7.47, 12.00]	2.8×10^{-8}	20/20
Main	MiniLM	Write–Choose	3.96 [2.81, 5.10]	7.0×10^{-7}	19/20
Main	OpenAI	Write–Choose	6.44 [4.88, 7.99]	5.0×10^{-8}	19/20
Length-matched	MiniLM	Write–Rewrite	1.94 [0.86, 3.02]	.0028	9/10
Length-matched	OpenAI	Write–Rewrite	2.35 [1.20, 3.51]	.0012	10/10
Length-matched	MiniLM	Write–Choose	1.15 [0.50, 1.80]	.0031	9/10
Length-matched	OpenAI	Write–Choose	1.48 [0.98, 1.99]	.000090	10/10

S.V.14 Upper bounds on dimensional increases from choosing and rewriting

We estimate how large a dimensional increase from choosing or rewriting is compatible with the data. Table S16 gives one-sided 95% upper confidence bounds across prompt groups. For comparison, it also gives the human excess under the same sampling conditions.

Table S16: Point estimates (mean ± 95% CI) and one-sided 95% upper bounds (UB) for the dimensional excess of choosing and rewriting, by analysis (MiniLM / OpenAI). The human mean excess is shown for scale. In the complete-sentence and sibling-free analyses, the point estimates for choosing are positive but not significant. No analysis shows a significant increase.

Analysis	Human mean	Choose mean ± CI	Choose UB	Rewrite mean ± CI	Rewrite UB
main	+2.30 / +3.69	−1.66±0.91 / −2.75±1.38	−0.91 / −1.61	−4.78±2.15 / −6.04±2.70	−3.00 / −3.81
length-matched (4-set)	+0.83 / +0.78	−0.32±0.93 / −0.71±0.79	+0.44 / −0.07	−1.10±1.11 / −1.58±1.05	−0.21 / −0.73
complete-sentence	+4.86 / +7.30	+0.92±0.94 / +1.00±1.57	+1.70 / +2.30	−0.58±2.45 / +0.46±3.45	+1.44 / +3.31
sibling-free	+5.05 / +7.72	+1.30±1.49 / +1.94±2.03	+2.53 / +3.62	−1.51±2.21 / −1.01±2.65	+0.32 / +1.18
menu-capped	+2.18 / +3.64	−1.45±1.00 / −2.27±1.54	−0.62 / −1.00	−4.50±2.25 / −5.51±2.82	−2.64 / −3.18
near-dup collapsed	+1.06 / +2.73	−2.65±0.88 / −3.70±1.25	−1.92 / −2.67	−5.19±1.89 / −6.36±2.46	−3.63 / −4.33

In the main, menu-capped, and near-duplicate analyses, the bounds exclude any dimensional gain from choosing or rewriting. Under the restricted references of the complete-sentence and sibling-free analyses, the bounds for choosing widen to +1.7–+3.6. But these bounds are below half of the human excess under the same conditions. Smaller gains are not excluded in these analyses.

S.V.15 Seed sensitivity

We reran the main analysis under seeds 41–45. Under MiniLM, the human dimensional excesses range from +2.23 to +2.52 (between-seed SD 0.11). Under OpenAI, they range from +3.69 to +4.04 (SD 0.13). For raw leakage, in the same model order, the excesses range from +0.0315 to +0.0320 (SD 0.0002) and from +0.0346 to +0.0352 (SD 0.0002). They are positive in 20 of 20 prompt groups at every seed under both embedding models.

S.VI Session Thirds

For story prompts (10 groups), Table S17 reports the raw-leakage and dimensional results for session thirds. Table S18 reports the orientation component that the main article uses, and Figure S1 plots it.

Table S17: Session-thirds results for story prompts (10 groups). Values are the excess over the control from the same session third. The equivalence column reports the original sensitivity analysis. Its margin is half of the early value. The text reports fixed-margin TOST results at ±0.020 and ±0.015. n.e. = not established.

	early	late	change	equivalence
Write, leakage (MiniLM)	+0.037 (p=.001)	+0.039 (p=5e-6)	+0.002 (p=.75)	**TOST** p=.023
Write, leakage (OpenAI)	+0.043 (p=6e-4)	+0.039 (p=1e-5)	−0.005 (p=.51)	**TOST** p=.014
Write, d_{eff} (MiniLM)	+2.18 (p=.034)	+2.83 (p=8e-4)	+0.65 (p=.53)	p=.33, n.e.
Choose, leakage (MiniLM)	−0.003 (n.s.)	−0.011 (p=.051)	−0.008 (p=.11)	—

The raw leakage of the first and final thirds is also equivalent when both genres are pooled (p=.046 and p=.015, 16 groups).

For the orientation component, we use the full analysis of all three session thirds. This analysis also includes Accepted sentences when it sets the common sample size. The mean excess stays positive but declines by about 0.006 under both embedding models (Table S18). Unadjusted two-sided paired t-tests give p=.094 / .048. Fixed-margin TOST gives $p \leq 4 \times 10^{-4}$ at ±0.020 and $p \leq .005$ at ±0.015. These tests exclude changes outside the stated margins. The ±0.020 margin is wide: it is close to the early values themselves (+.025 / +.028). For this reason, we also report the narrower ±0.015 margin and margins of half the early value, which also show equivalence (p=.018 / .0099). None of these tests establishes an absence of decline. Each estimate uses the model reference from the same session third. In this way, it adjusts for changes in that reference over the session.

Table S18: Writing's orientation component by session third (10 story prompt groups). The paired change is late minus early. Its 90% Student t interval supports the two one-sided tests at the 5% level. The reported two-sided p-values are unadjusted.

Model	Early	Late	Change	90% CI for change	p
MiniLM	+.0252	+.0197	−.0055	[−.0108, −.0001]	.094
OpenAI	+.0277	+.0215	−.0062	[−.0111, −.0012]	.048

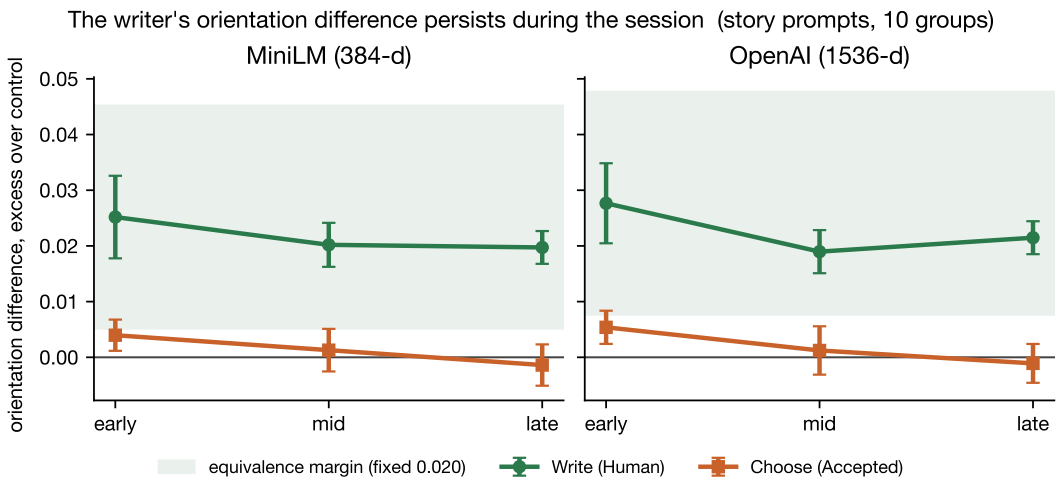


Figure S1: Orientation difference (orientation component) in each third of a session. Each value is measured against the model reference from the same session third. The band is the fixed ±0.020 equivalence margin around the early value. Human-dominant sentences show a small mean decline that stays within this margin. Accepted sentences also show a downward tendency.

S.VII Residual Spectrum

Figure S2 supports the Discussion in the main text. For each prompt group, we removed the model's top-20 subspace from the human sentences and pooled the residuals. As a control, we did the same for the held-out half of the dismissed pool. This control contains no human text. The leading component of the human residual carries 0.85% of residual variance on story prompts and 0.92% on opinion prompts. For the control, the values are 0.76% and 0.85%. An isotropic distribution of the same dimension would give about 0.066%. The leading component is therefore about 13–14 times this level, so the residual is not isotropic. Still, the concentration on a few axes is limited: together, the top three human components carry 2.5% and 2.7%. The human spectrum is also close to that of the control.

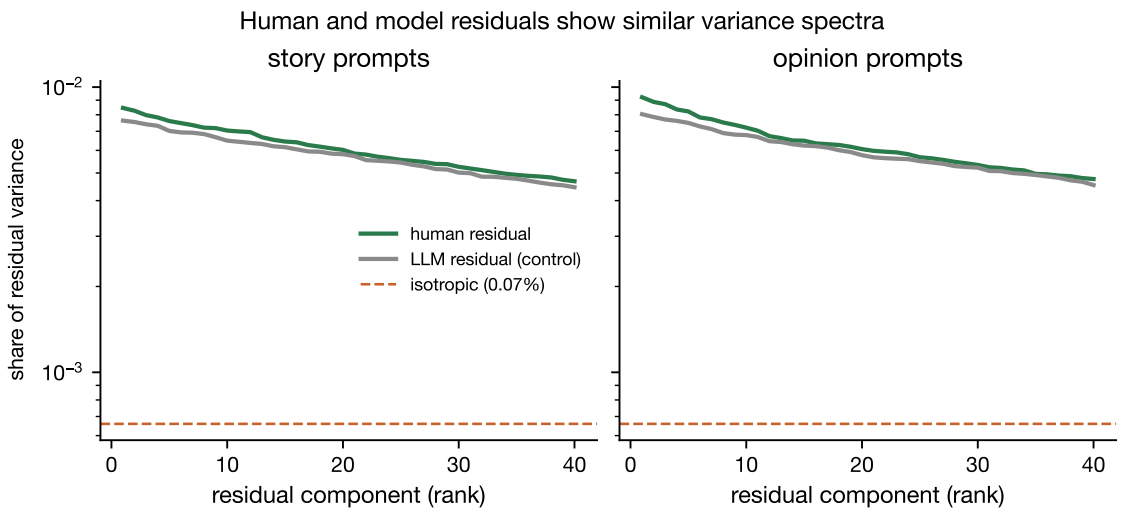


Figure S2: Residual variance spectra after projecting out the model's top-20 subspace, pooled across prompt groups (text-embedding-3-small, 1,536 dimensions). The human residual and the LLM-residual control have nearly the same spectrum. The dashed line is an approximate isotropic level, 1/1516 (1,536 minus the 20 removed dimensions). It is approximate because each prompt group removes a different subspace before pooling.

S.VIII Engagement Moderators

Table S19 and Figure S3 report the full moderator analysis. This analysis is exploratory and is reported here only. Within each prompt group, we split sessions into quartiles of three behavioural moderators. We then measured the human sentences of each quartile using equal-sized samples. Writers who reject more suggestions also feed more sentences into the reference pool. For this reason, each repetition splits the group's sessions in half. In this way, no session supplies both the reference and the sentences measured against it.

Table S19: Trend of the writer's raw leakage across the quartiles of each moderator.

moderator	Q1 → Q4	groups	leakage trend (MiniLM / OpenAI)
dismissal rate	.70 → .89	20	−0.0025 (p=.024) / −0.0016 (p=.095)
editing rate	.07 → .88	14	−0.0041 (p=.0008) / −0.0019 (p=.064)
own-writing share	.36 → .90	17	−0.0013 (p=.10) / −0.0008 (p=.059)

This analysis found no evidence that any moderator predicts a larger raw-leakage excess. Instead, the trends are slightly negative, and two of them reach $p < .05$ under MiniLM. Raw leakage lies between +0.031 and +0.051 in all twelve moderator-by-quartile cells. Every cell has $p < .001$. In the weakest cell, it is positive in all prompt groups but one.

Two caveats limit this result. First, dismissal rates span only .70 to .89, so writers who mostly accept what they are offered are barely represented. Second, quartiles pool sessions, so variation between individual writers is partly averaged away.

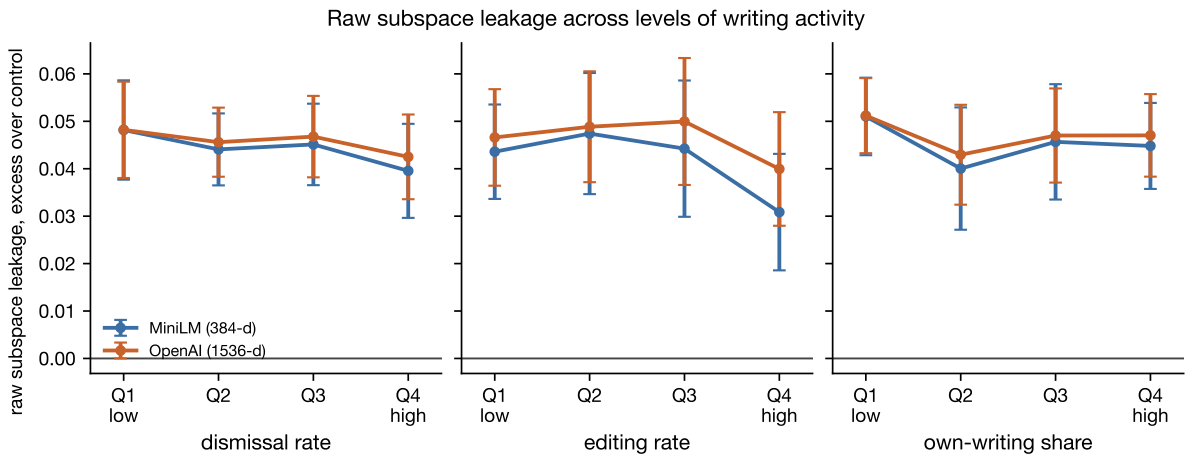


Figure S3: The writer's raw subspace leakage (ΔL_{20}) by quartile of three engagement moderators, for both embedding models. Here the reference is session-disjoint, so levels are not comparable with the main analysis. No panel shows a rising trend; some show a slight decline.

S.IX GPT-5 Regeneration Details

We replayed 2,275 contexts from 300 sessions. We used 15 sessions per prompt group, with up to ten evenly spaced suggestion requests in each. For each context, we collected five GPT-5 continuations. Of the 11,375 requested continuations, 11,308 were kept after we filtered out empty and truncated outputs. No request failed.

The regenerated pool keeps CoAuthor's sibling structure (five continuations per context), so menu-level redundancy is parallel by construction. The API for this model does not expose a sampling temperature. Its decoding settings are therefore the provider defaults, while CoAuthor used 0.30 and 0.75. The continuations were capped at 80 tokens, so the length distribution of the regenerated pool differs from CoAuthor's. The GPT-5 pool also draws on fewer contexts per prompt group than the original.

Effective dimension with new samples. We subtract the same held-out GPT-5 control from Human and from GPT-3. This subtraction does not make their effective-dimension contrast depend on the geometry of that control:

$$[d_{\text{eff}}(H) - d_{\text{eff}}(C_5)] - [d_{\text{eff}}(G_3) - d_{\text{eff}}(C_5)] = d_{\text{eff}}(H) - d_{\text{eff}}(G_3),$$

where H, G_3, and C_5 denote equal-sized samples of human text, original model suggestions, and the held-out GPT-5 control. The new pool changes the available sample sizes. Under length matching, it also changes the texts selected. So this analysis tests the human–original-model difference after we adjust sample sizes. In the length control, it also adjusts sentence lengths. It does not independently test that difference against the geometry of GPT-5 text. Before length matching, the mean difference is +1.83 under MiniLM and +2.92 under OpenAI. The sign and ordering replicate across embedding models, but the magnitudes differ.

Raw leakage and its orientation component. Human and original model text have equivalent raw adjusted leakage within the tested ±0.016 margin (TOST $p \leq 5.2 \times 10^{-6}$). This composite measure mixes spectral shape and orientation, so its equivalence does not establish equivalent orientation. The decomposition gives separate evidence. All four original-corpus sets have positive orientation excesses relative to the GPT-5 reference (+.091 to +.116, 20/20 groups for each set under both embedding models). The mean is +.0906 / +.0936 for human text and +.1032 / +.1045 for original model suggestions. The paired human–original-model differences are −.0126 / −.0109 (Table S20). In this analysis, the human orientation component is therefore lower. This result differs from its positive excess over the original model reference. The analysis uses equal-sized samples but does not match sentence lengths.

Effective dimension of combined collections. The union analysis directly includes GPT-5 text in each collection. It compares $d_{\text{eff}}(R_5 \cup X)$ with $d_{\text{eff}}(R_5 \cup C_5)$. Here, R_5 and C_5 are disjoint GPT-5 samples. The set X contains the same number of original model or human sentences. For original model text, the mean increase is +4.78 / +10.18 (18/20 and 20/20 groups positive). For human text, it is +5.36 / +11.56 (20/20 under both embedding models). On average, both sources increase the effective dimension of the combined collection. The direct human–original-model difference is +0.58 / +1.38. However, the MiniLM interval includes zero (Table S20). So the mean human increase is larger under both embedding models, but evidence for the direct difference is clear only under OpenAI. This union analysis does not include length matching.

The pools also differ sharply in length. For the GPT-3 suggestions, the word counts are p10/p50/p90 = 6/12/23. They are 19/27/37 for the regenerated GPT-5 suggestions and 7/14/26 for human sentences. We therefore applied a three-set length match to the reference, GPT-3, and Human sets (4-word bins; 20 groups; n 49–81 per set). This match removes the apparent pool concentration under MiniLM (GPT-3 − GPT-5-control +0.09, p=.91, 11/20). It halves the concentration under OpenAI (+5.20, p=2.2e-4, 16/20). The human–original-model dimensional difference remains positive (+1.27, p=.013, 17/20 MiniLM; +1.56, p=.0067, 15/20 OpenAI). But their raw-leakage difference remains near zero. The main text therefore limits the pool comparison to the tested sampling and decoding settings. It does not treat the comparison as a general property of model generations.

Table S20: Direct human–original-model contrasts in the GPT-5 analyses (20 prompt groups). Intervals are paired 95% Student t intervals. Two-sided p-values are unadjusted. Positive counts refer to prompt groups.

Analysis	Model	Mean [95% CI]	p	Positive
Dimension, length-matched	MiniLM	$+1.27$ $[+.30, +2.23]$	.013	17/20
Dimension, length-matched	OpenAI	$+1.56$ $[+.49, +2.64]$	.0067	15/20
Orientation component	MiniLM	$-.0126$ $[-.0162, -.0091]$	5.1×10^{-7}	2/20
Orientation component	OpenAI	$-.0109$ $[-.0142, -.0075]$	1.8×10^{-6}	2/20
Union dimension	MiniLM	$+.58$ $[-.10, +1.26]$	.091	12/20
Union dimension	OpenAI	$+1.38$ $[+.59, +2.18]$	.0017	16/20

The dimensional length match uses GPT-5, original model, and human pools, with n = 49–81 per set. The orientation and union analyses use n = 117–120 and do not match lengths. Sampling conditions and measures differ, so magnitudes cannot be compared directly across analyses.

Direct comparison with the regenerated reference. Without length matching, human sentences exceed the GPT-5 control in effective dimension by +4.39/+13.1 (13/20 and 20/20 groups positive). The original suggestions also exceed it (+2.56/+10.2), so the regenerated pool is narrower under these sampling and decoding settings. After three-set length matching, the direct human–GPT-5 difference is +1.36 under MiniLM (14/20, p = .15) and +6.77 under OpenAI (17/20). The human–original-model ordering therefore persists. However, a reliable human excess over the regenerated reference does not hold under both representations. Removing empty and truncated generations also favours complete sentences. The original writers never interacted with these regenerated suggestions. For this reason, none of these comparisons is a new co-writing experiment.

S.X Writing Process and Proxy-Matching Checks

This section describes the writing-process and proxy-matching checks that the Results of the main text summarize. The code has four scripts: `process_flags.py` (flags), `process_delta.py` (same-condition comparison), `process_robust.py` (variants), and `coherence_check.py` (proxy matching). All scripts use seed 42.

S.X.1 Process flags

The first process flag tracks deleted suggestion text. The main replay (see Method in the main text) tags every inserted character with its source (writer or suggestion) and its event index. It keeps only the characters that survive to the final document. The flag replay processes the same delta stream with one addition. When a delete operation removes characters, we count the removed suggestion characters, plus any marks those

characters carried. We add this count to a counter on *both* surviving neighbours of the deletion point. Marks move with their characters under later edits. A sentence's `ai_ghost` value is the sum of marks on its surviving characters. Because both neighbours are marked, a deletion at a sentence boundary marks both sentences, so the flag over-marks rather than misses.

The bigram overlap of a sentence is the share of its word pairs that already appeared in earlier suggestions. To compute it, we lower-case the text and tokenize it with the pattern `[a-z0-9']+`. We then take the set of adjacent word pairs in the sentence. We compare this set with the union of word pairs from every suggestion shown earlier in the session (`suggestion-open` events, using the trimmed text). "Earlier" means at an event index smaller than that of the sentence's last surviving character. So a sentence is treated as finished when its last kept character was typed. The overlap is the share of the sentence's pairs that are found in that union. As a validity check, the median overlap is 1.00 for accepted suggestions, 0.55 for rewritten sentences, and 0.05 for human-written sentences. Of 21,267 human-written sentences, 1.8% carry a deletion mark, 2.4% contain a surviving suggestion character, and 8.1% have an overlap of at least 0.30.

The 0.30 threshold was fixed before the geometric analyses were run. It is a round value well above the human median and well below the rewritten median. We also report 0.20 as a stricter check. Short sentences built from common phrases (for example, "You're going to be okay.") can reach high overlap by chance. For this reason the filter is conservative: it removes some sentences that were not derived from suggestions. The flags cannot detect a suggestion whose idea was adopted in different words.

S.X.2 Same-condition comparison

This comparison tests both human sets against one shared control sample. In each prompt group and each of 20 replicates, we split the dismissed pool in half at random. We draw one held-out control sample of size n from one half. Then we draw two human samples of the same size. One comes from all human-written sentences, and the other comes from the process-filtered sentences. Both effective-dimension excesses are computed against the same control sample. All draws are without replacement. Table S21 lists the pools and the per-group differences. We report the mean difference (process-filtered minus all) and its 95% t interval across prompt groups. We also give the number of groups with a positive difference. At threshold 0.30, the difference is $+0.58$ $[+0.26, +0.91]$ (16/20) under MiniLM and $+0.89$ $[+0.52, +1.26]$ (18/20) under OpenAI. At 0.20, it is $+1.06$ $[+0.67, +1.46]$ (18/20) under MiniLM and $+1.39$ $[+0.90, +1.87]$ (18/20) under OpenAI.

Table S21: Per-group pools and same-condition differences in the effective-dimension excess. The difference is process-filtered minus all human-written sentences, with overlap threshold 0.30. N_H: human-written sentences available; N_P: process-filtered sentences; n: size of every sample.

Prompt group	N_H	N_P	n	MiniLM	OpenAI
opinion 1	808	724	200	+0.30	+0.73
opinion 2	693	638	200	+0.98	+1.14
opinion 3	708	646	200	+1.94	+1.78
opinion 4	696	637	200	+0.86	+0.62
opinion 5	712	649	200	+0.40	+0.17
opinion 6	661	588	200	+0.45	+0.60
opinion 7	694	648	200	+0.26	+0.58
opinion 8	679	610	200	+1.12	+1.47
opinion 9	785	718	200	+1.16	+2.29
opinion 10	784	721	200	−0.07	−0.01
story 1	1,399	1,242	200	−0.97	−0.29
story 2	1,498	1,338	200	+1.09	+1.31
story 3	1,206	1,040	200	−0.23	+0.95
story 4	1,423	1,272	200	−0.12	+0.41
story 5	1,333	1,156	200	+0.57	+1.55
story 6	1,300	1,138	200	+0.73	+1.07
story 7	1,157	984	200	+1.96	+2.78
story 8	1,590	1,400	200	+0.52	+0.19
story 9	1,746	1,589	200	+0.43	+0.26
story 10	1,395	1,266	200	+0.29	+0.19

Table S22: Effective-dimension contrasts under process-based definitions of writing. W = written, C = held-out control, R = rewritten. Entries are mean paired differences across prompt groups, with 95% t intervals and the number of groups with a positive difference. N_H: human-written sentences available; n: sample size per set (range over groups). In the last variant, excluded human sentences join the rewritten set. This enlarges that pool, so more prompt groups pass the length-matched minimum. Each row draws its samples independently. Table S21 gives the same-condition comparison.

Human definition	Match	Contrast	N_H	n	MiniLM	OpenAI
Character origin (main)	none	W−C	21,267	141–200	+2.30 [+1.05, +3.55] (15/20)	+3.69 [+1.86, +5.52] (18/20)
Character origin (main)	none	W−R	21,267	141–200	+7.08 [+5.20, +8.95] (20/20)	+9.74 [+7.47, +12.00] (20/20)
Process-filtered, overlap < 0.30	none	W−C	19,004	141–200	+2.75 [+1.52, +3.98] (17/20)	+4.37 [+2.49, +6.25] (19/20)
Process-filtered, overlap < 0.30	none	W−R	19,004	141–200	+7.53 [+5.72, +9.33] (20/20)	+10.41 [+8.20, +12.62] (20/20)
Process-filtered, overlap < 0.20	none	W−C	16,878	141–200	+3.13 [+1.81, +4.46] (16/20)	+4.83 [+2.88, +6.77] (19/20)
Process-filtered, overlap < 0.20	none	W−R	16,878	141–200	+7.91 [+6.18, +9.64] (20/20)	+10.87 [+8.72, +13.02] (20/20)
Overlap < 0.30 and $h = 1$	none	W−C	18,751	141–200	+3.20 [+1.92, +4.48] (19/20)	+4.85 [+2.86, +6.85] (19/20)
Overlap < 0.30 and $h = 1$	none	W−R	18,751	141–200	+7.98 [+6.16, +9.80] (20/20)	+10.90 [+8.61, +13.19] (20/20)
Excluded Human → rewrite	none	W−C	19,004	141–200	+2.74 [+1.47, +4.02] (16/20)	+4.39 [+2.50, +6.28] (19/20)
Excluded Human → rewrite	none	W−R	19,004	141–200	+6.56 [+5.32, +7.80] (20/20)	+8.98 [+7.35, +10.61] (20/20)
Character origin (main)	length	W−C	21,267	40–62	+0.83 [+0.00, +1.66] (7/10)	+0.78 [+0.12, +1.44] (8/10)
Character origin (main)	length	W−R	21,267	40–62	+1.94 [+0.86, +3.02] (9/10)	+2.35 [+1.20, +3.51] (10/10)
Process-filtered, overlap < 0.30	length	W−C	19,004	40–62	+0.87 [+0.02, +1.72] (8/10)	+0.78 [+0.07, +1.49] (8/10)
Process-filtered, overlap < 0.30	length	W−R	19,004	40–62	+1.98 [+0.70, +3.25] (9/10)	+2.36 [+1.04, +3.67] (9/10)
Process-filtered, overlap < 0.20	length	W−C	16,878	40–62	+1.15 [+0.23, +2.06] (9/10)	+1.11 [+0.42, +1.80] (9/10)
Process-filtered, overlap < 0.20	length	W−R	16,878	40–62	+2.25 [+1.05, +3.45] (9/10)	+2.69 [+1.62, +3.75] (10/10)
Overlap < 0.30 and $h = 1$	length	W−C	18,751	40–62	+0.94 [+0.19, +1.69] (9/10)	+0.88 [+0.25, +1.51] (8/10)
Overlap < 0.30 and $h = 1$	length	W−R	18,751	40–62	+2.05 [+0.87, +3.22] (9/10)	+2.46 [+1.13, +3.79] (9/10)
Excluded Human → rewrite	length	W−C	19,004	40–95	+1.52 [+0.72, +2.31] (15/17)	+1.86 [+0.90, +2.82] (14/17)
Excluded Human → rewrite	length	W−R	19,004	40–95	+2.77 [+2.04, +3.51] (16/17)	+3.06 [+2.14, +3.99] (16/17)

S.X.3 Document similarity and GPT-2 log-perplexity

This subsection defines the two proxies used for matching. Document similarity is the cosine similarity between a sentence and the centroid of its session's final-document sentences (human, accepted, and rewritten). The sentence itself is excluded from the centroid. Dismissed suggestions are compared with the same centroid. We exclude sessions with fewer than three document sentences. This exclusion slightly changes the unmatched estimate. Log-perplexity is the mean negative log-likelihood per token under GPT-2 (124M), with a beginning-of-text token prepended. Both measures are proxies, not human ratings. The final document consists largely of the writer's own sentences. So human sentences are compared with a centroid partly built from the same writer's text, which favours them on similarity.

We first report the median of each proxy. For human-written sentences, median similarity is 0.491 (MiniLM) / 0.507 (OpenAI), compared with 0.483 / 0.493 for dismissed suggestions. Median log-perplexity is 4.00 for human-written sentences and 3.67 for dismissed suggestions. Next, we compute quintile cut points of both proxies on the dismissed reference, which gives 5×5 joint bins. Within each prompt group, we subsample human-written sentences without replacement so that their bin shares equal those of the reference. The most constrained bin sets the size of the human pool, while the reference is left unchanged. The geometry pipeline then draws equal-sized human and control samples without replacement. This matching is separate from the length matching described in the Method of the main article. No sentence appears twice in any sample.

Table S23: Human minus control effective dimension before and after matching document similarity and GPT-2 log-perplexity to the reference. Entries give the mean, 95% t interval, and groups positive. n: sample size per set (range over groups). Magnitudes are not comparable across rows because n differs.

Human set	n (MiniLM)	MiniLM	n (OpenAI)	OpenAI
Unmatched	200–200	$+2.21$ $[+0.86, +3.56]$ (15/20)	200–200	$+3.98$ $[+2.02, +5.94]$ (18/20)
Matched to reference	65–200	$+1.64$ $[+0.61, +2.67]$ (16/19)	110–200	$+3.98$ $[+2.47, +5.49]$ (17/17)

S.XI Fixed-Design and Session-Contribution Checks

These exploratory checks separate the effect of length restriction from changes in sample size and prompt composition. They use the saved sentence embeddings, seed 42, and 20 draws per prompt, and they introduce no new model outputs or annotations. The released `sampling_sensitivity.py` script and its CSV files report all four settings below. They also report all three direct writing contrasts, prompt-level estimates, and session-contribution diagnostics.

S.XI.1 Length restriction at fixed prompts and sample sizes

This check holds the prompts and sample sizes fixed. We keep exactly the ten prompts and the per-prompt sizes ($n = 40$–62) of the original four-set length-matched analysis. The unrestricted setting draws from the original pools. By contrast, the restricted setting first selects equal counts across Accepted, Mixed, Human, and Dismissed in each four-word bin. It uses the original algorithm and random draws. Drawn samples can still differ in their realized bin counts. This is because the restriction balances the candidate pools but does not ensure identical lengths within bins. Both settings split the reference and draw equal-sized sets. We compute the participation ratio from the trace of the centred Gram matrix, without a full singular-value decomposition. This calculation agrees with the saved length-matched estimates within 6×10^{-6} under both encoders.

Table S24: Writing minus rewriting at the same ten prompts and per-prompt sizes. Entries are means [95% t intervals across prompts]. The number of positive prompts appears in parentheses. The intervals rest on a working assumption that prompts are independent. They are not cluster-calibrated.

Sampling setting	MiniLM	OpenAI
Unrestricted	2.50 [1.22, 3.78] (10/10)	2.96 [1.70, 4.22] (10/10)
Length-restricted	1.94 [0.86, 3.02] (9/10)	2.35 [1.20, 3.51] (10/10)
Session-weighted	3.66 [1.25, 6.07] (9/10)	4.41 [1.91, 6.91] (10/10)
Length + session-weighted	3.01 [1.04, 4.99] (9/10)	3.86 [1.66, 6.06] (9/10)

Length restriction reduces the fixed-design gap by 0.56/0.60 effective dimensions under MiniLM/OpenAI. This is about 22%/20% of its unrestricted mean. These ratios are descriptive and are not estimates of the causal contribution of length. Comparing the original main gap (7.08/9.74) with 1.94/2.35 would instead mix length restriction with different prompts and sample sizes. Even within the fixed design, restricting the candidate pools can change content and session composition as well as length.

S.XI.2 Unequal session and writer contributions

This check describes how much each session and writer contributes. For each condition and prompt, we count the distinct source sessions and their shares of candidate sentences (Table S25). In story prompts, writer counts use the metadata identifiers, but opinion prompts have no writer identifiers in the metadata file we used. These counts describe candidate pools, not independent observations or effective sample sizes. Within story prompts, the largest writer contributes 14.5–24.3% of rewritten sentences, compared with 7.1–10.8% of predominantly human-written sentences. So equal sentence counts do not imply an equal mix of contributors.

Table S25: Contribution ranges across prompt-specific candidate pools. Session ranges cover all 20 prompts, while writer ranges cover the ten story prompts. The largest-session share is the share of the largest contributor within each pool.

Pool	Sessions	Largest session share	Story writers
Dismissed	52–90	2.8–9.2%	33–39
Accepted (choosing)	45–87	3.7–13.0%	32–39
Mixed (rewriting)	43–82	3.5–10.7%	24–36
Human (writing)	56–90	2.4–5.0%	33–39

As a sensitivity check for composition, we reweight sentences by session. Each sentence gets an initial sampling weight that is inverse to the number of sentences its session has in the relevant candidate pool. We then draw without replacement. Reference weights are calculated within each random split half. The weighting gives less weight to sessions that contribute many sentences, but it does not enforce equal realized session counts. It also does not reproduce a cluster bootstrap or establish interval coverage. We repeat this setting with and without length restriction, at the same prompts and n.

The writing–rewriting difference remains positive on average in all settings, but its magnitude changes (Table S24). This supports persistence under these composition changes, but it does not support invariance to clustering. Weighting also changes the length distribution within the restricted pool, so the combined setting is not an exact joint match on length and session. The machine-readable output keeps all contrasts, including writing–reference results that are less stable. Stronger inference still requires independent writers and prompts, as well as validation of a suitable resampling procedure for this finite-sample statistic.